

EXHIBIT 45



HDK8450 (SM8450P) Processor

User Guide

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December 2021

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Revision history

Revision	Date	Description
A	September 2021	Initial release
B	September 2021	Optimize mainboard top/bottom items list sequence
C	November 2021	Updated Figure 3-1 . Updated Figure 3-3 . Updated Figure 3-4 .

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1 Introduction

1.1 Purpose

This document provides a quick reference or evaluation platform for Qualcomm's latest 8450 series - Snapdragon 8450 processor. This is suited for Android/Linux application developers, OEMs, consumer manufacturers, hardware component vendors, video surveillance, robotics, camera vendors, and flash chip vendors to evaluate, optimize, test and deploy applications that can utilize the Qualcomm Snapdragon 8450 series technology.

1.2 Conventions

Function declarations, function names, type declarations, attributes, and code samples appear in a different font, for example, `#include`.

Code variables appear in angle brackets, for example, `<number>`.

Commands to be entered appear in a different font, for example, **copy a:*. * b:**.

Button and key names appear in bold font, for example, click **Save** or press **Enter**.

Shading indicates content that has been added or changed in this revision of the document.

1.3 Technical assistance

For assistance or clarification on information in this document go to: <https://tech.intrinsyc.com>

2 System block diagram

The HDK8450 development platform consists of two major components:

- Main board
- Display card (Display panel + Expansion card)

The following diagrams show the top and bottom on the development platform.

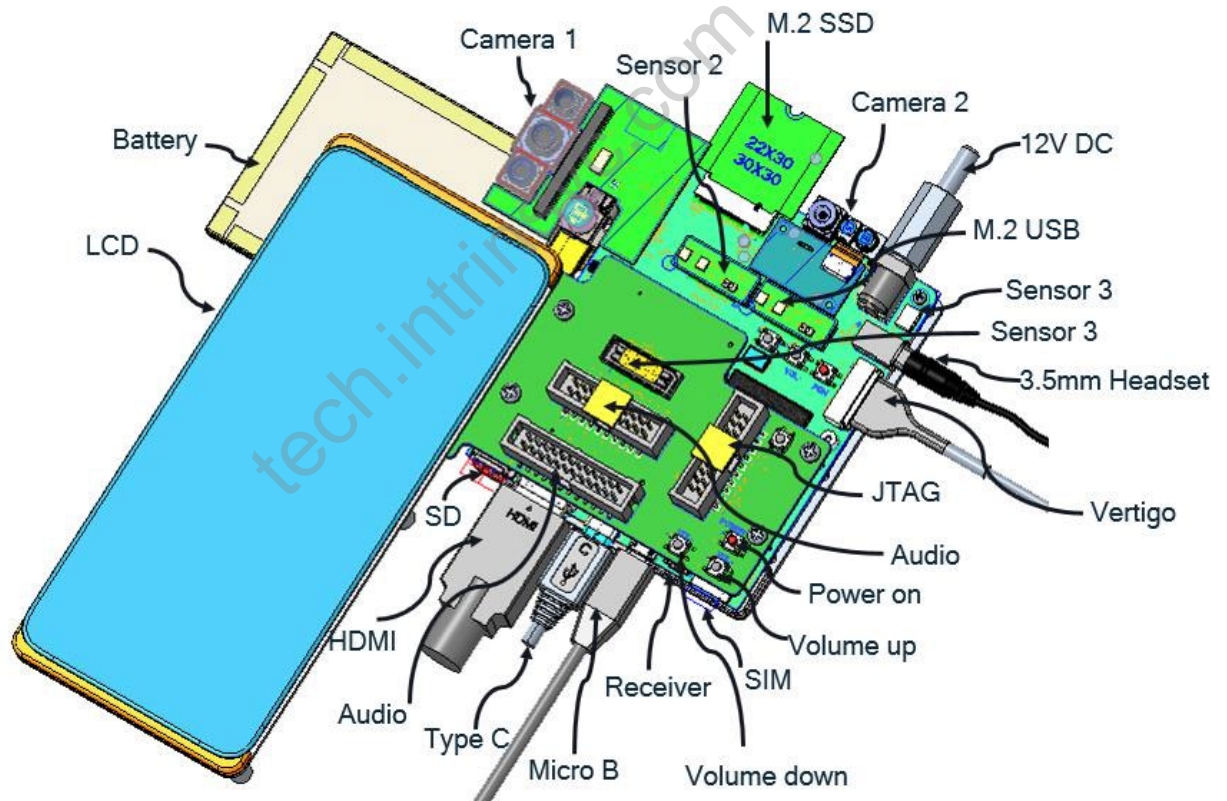


Figure 2-1 HDK8450 platform top

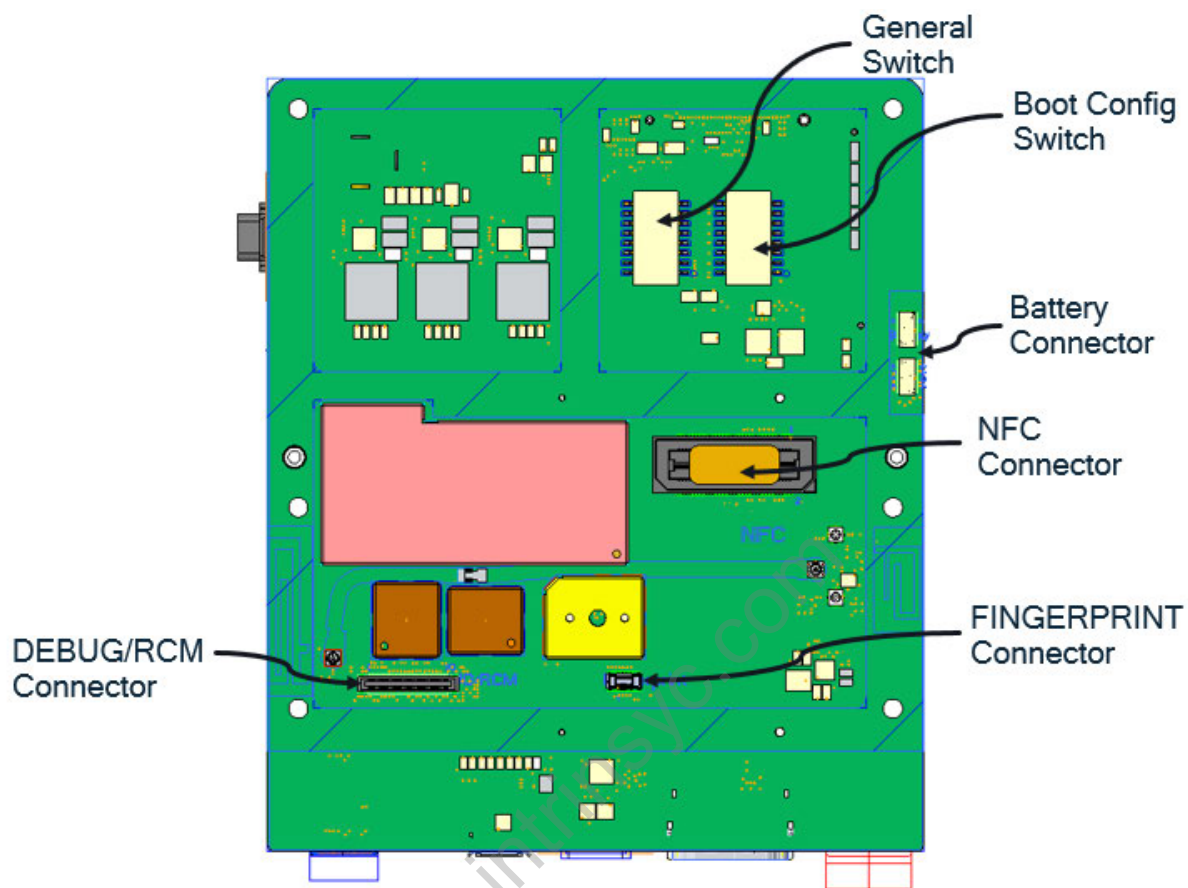


Figure 2-2 HDK8450 platform bottom

The following diagram explains the interconnectivity and peripherals on the development kit.

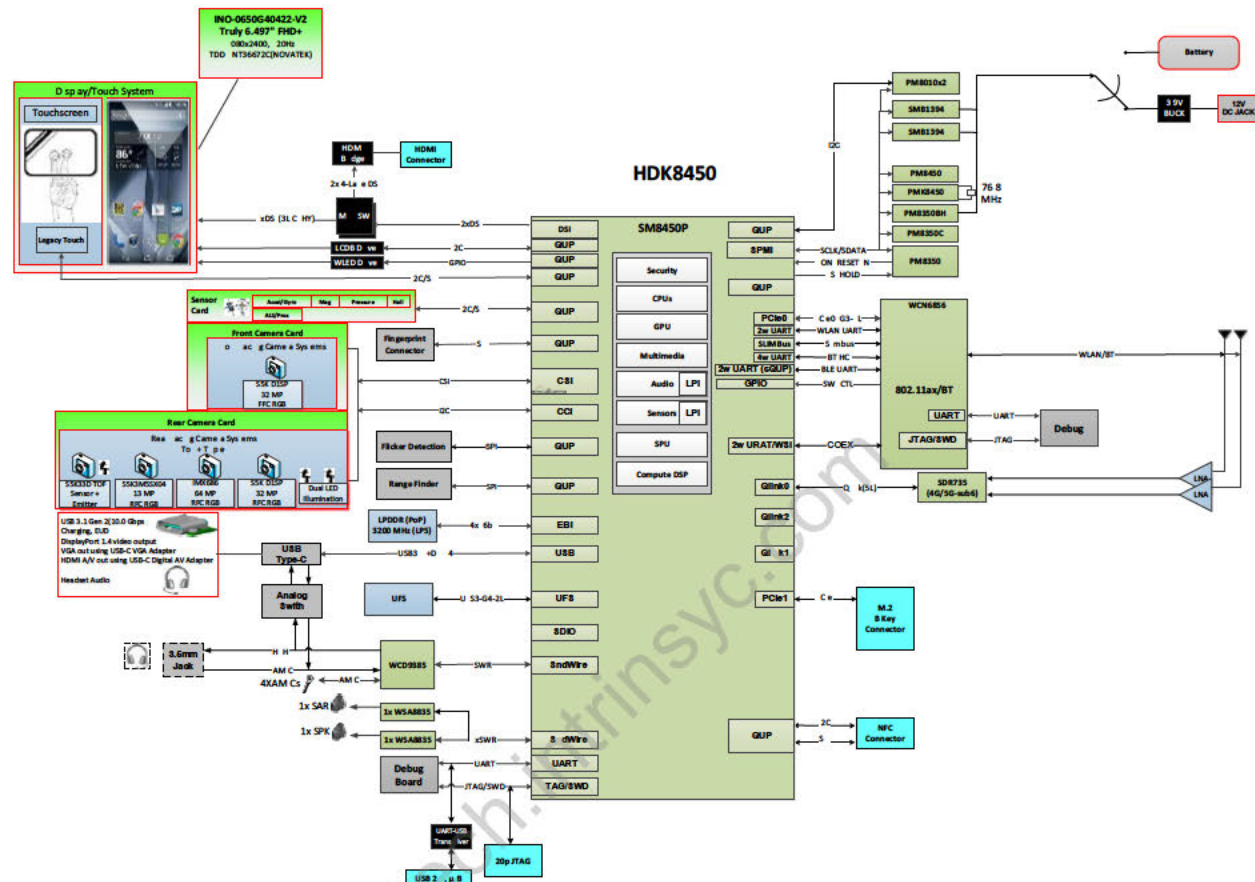


Figure 2-3 HDK8450 block diagram

3 Main board

The main board provides the basic common set of features with minimal integration efforts for end users. It contains the followings:

- Snapdragon 8450 (SM8450P) main application processor
- Memory: 256 GB UFS 3.0 + LPDDR5 up to 3200 MHz 12 GB RAM
- PMIC: PMK8450+ PM8450 + PM8350 + PM8350C + PM8350B + PM8010 + PMR735A
- WCN6856 Wi-Fi + BT combo chip over SLIMbus, PCIE, UART, PCM
- WCD9385 Audio Code
- SDR735

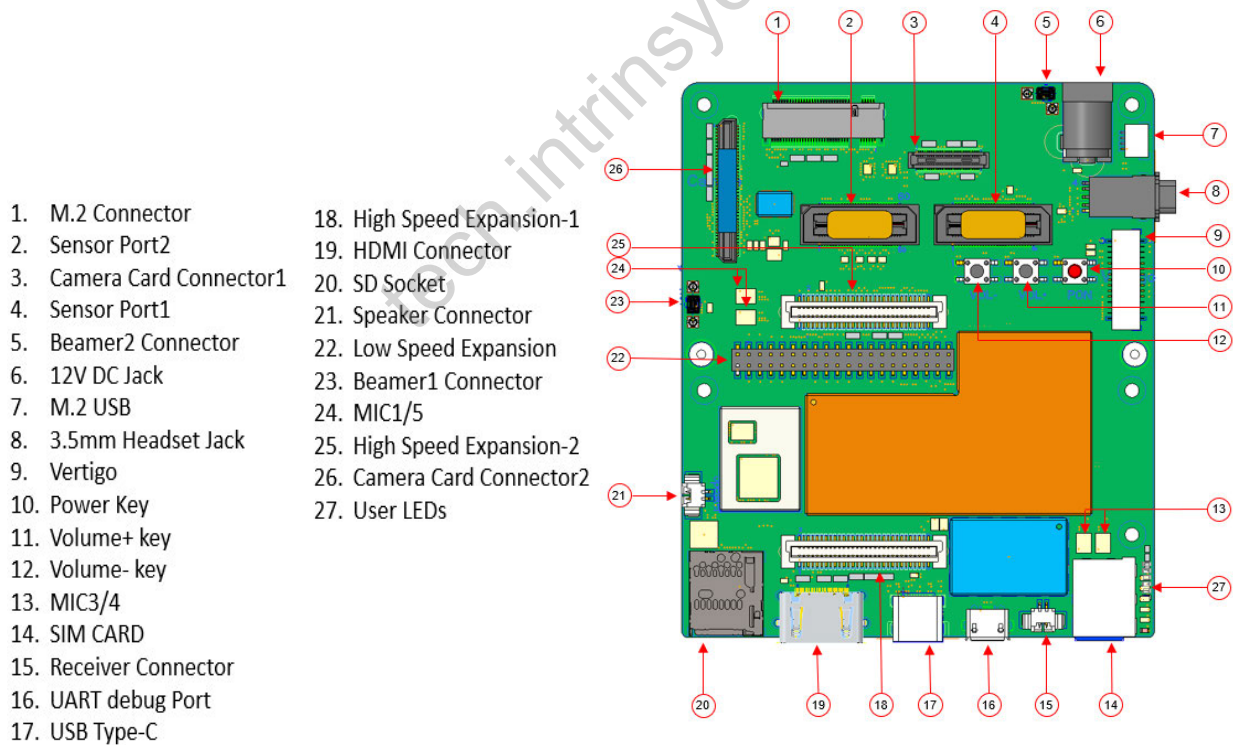


Figure 3-1 HDK8450 main board top

- 28. General Switch(S5702)
- 29. Boot Config Switch(S5701)
- 30. Battery Connector
- 31. NFC Connector
- 32. Wi-Fi CH0 Antenna
- 33. Wi-Fi CH1/GNSS Antenna
- 34. Fingerprint Connector
- 35. Debug/RCM Connector

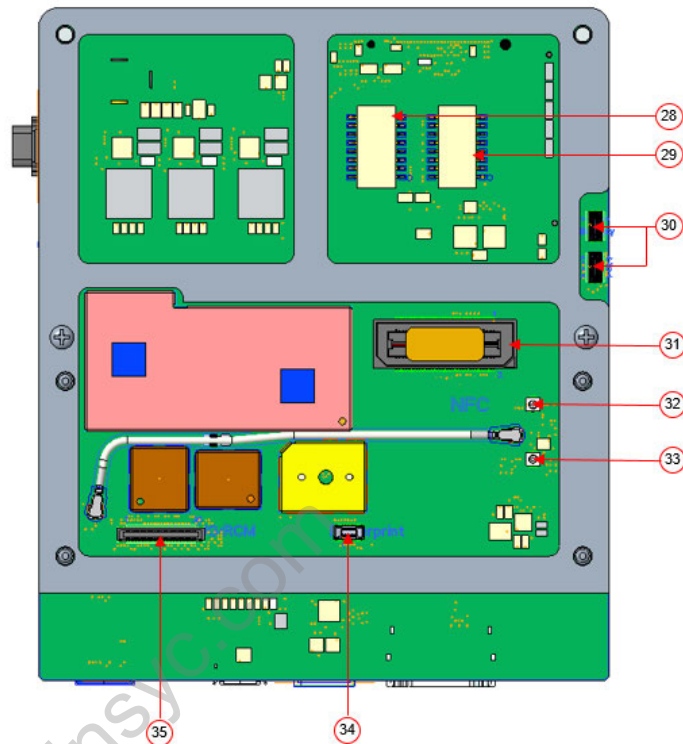


Figure 3-2 HDK8450 main board bottom

3.1 Main board mechanical properties

The HDK8450 Main Board measuring 100 mm x 85 mm is where all the processing occurs. It is connected to the Expansion Card via two 60 pin high speed, and one 40 pin low speed board-to-board connectors. The purpose of the connectors is to bring out essential signals so that other peripherals can interface with the platform.

Table 3-1 HDK8450 MAIN BOARD Mechanical Properties

Dimension	85 cm ² (100 mm x 85 mm)
Interface	2 x 60 pin high speed and 1 x 40 pin low speed board-to-board connectors
Thermal	A top side graphite sheet is stuck by default.

3.2 Hardware specification

Table 3-2 HDK8450 Hardware Features

Subsystem / Connectors	Feature Set	Description	Specification
Chipset	SM8450P	Qualcomm® Snapdragon™ 8450 Processor	- Qualcomm® Kryo™ 780 CPU built on Arm Cortex technology - Qualcomm® Adreno™ 730 GPU for the highest in graphics performance and power efficiency - Qualcomm Spectra™ 680 Image Signal Processor for the ultimate photography and videography experiences - Adreno 665 VPU for high-quality, ultra HD video encode and decode - Adreno 1195 DPU for on-device and external ultra HD display support 3G/4G/5G modem – mmWave and sub-6 GHz bands (Rel 16 integrated modem) - Low-power audio subsystem combined with the Qualcomm Aqstic™ Audio Technologies WCD9380/WCD9385 audio codec for low-power voice processing and audiophile quality audio playback - Qualcomm® Snapdragon Sensors Core Technology for contextual awareness and always-on sensor support - Qualcomm® Secure Processing Unit (SPU260) for advanced secure use cases - Qualcomm® Hexagon™ Tensor Processor (HTP) with Hexagon Vector eXtensions (HVX) and Hexagon Matrix eXtensions (HMX) - Qualcomm® FastConnect™ 6900 System with WCN685x, 802.11ax, 2 × 2 MIMO, Bluetooth 5.2 - Quad-channel package-on-package (PoP) high-speed LPDDR5 SDRAM 14.0 × 15.0 mm MPSP1518 PoP
	Qualcomm® PMIC, Companion PMIC for SM8450P processor (PMK8450, PM8450, PM8350, PM8350C, PM8150B, PM8010 and PM735A)	PMK8450	The PMK8450 device serves as a clock generation and distribution IC. - Three dedicated low dropouts (LDOs) for clocks - One for the crystal oscillator - One for the RFCLK output buffers - One for the LNBB output buffers 38.4/76.8 MHz XO support - Special-purpose clock outputs - Five 76.8/38.4 MHz RF clock outputs - Three 19.2 MHz LNBB clock output - One 32.7645 kHz sleep clock output - Two-line serial power management interface (SPMI) - On-chip ADC - Real time clock (RTC) 4 GPIOs (1.8 V) - Package 36-pin fanout wafer picoscale package (FOWPSP36) 2.13 × 2.13 × 0.55 mm; 0.35 mm pitch

Subsystem / Connectors	Feature Set	Description	Specification
		PM8450	The PM8450 device integrates most of the wireless product's power management, general housekeeping, and IC-level interface support functions into a single mixed-signal IC. - PM8450 is optimized for the SM8450 chipset. - Six FTS520 SMPS - Four LDO512 linear regulators - Overtemperature protection - SPMI - Four GPIOs 81-pin fan-out wafer-level pico-scale package (81 FOWPSP)
		PM8350	The PM8350 device integrates most of the wireless product's power management, general housekeeping, and IC-level interface support functions into a single mixed-signal IC. - PM8350 is optimized for the SM8450 chipset. - Three HFS510 SMPS and nine FTS520 SMPS 10 LDO512 linear regulators - On-chip ADC - Overtemperature protection - SPMI 10 GPIOs 144-pin fan-out wafer-level nano-scale package (144 FOWNSP)
		PM8350C	The PM8350C PMIC supplements the core PM8350 PMIC and the slave PM8350B PMIC to integrate all wireless handset power management, general housekeeping, IC-level interface, and user interface support functions into an integrated solution. - PM8350C is optimized for the SM8450 chipset. - Nine FTS520 SMPS and one HFS515 SMPS regulators 13 LDO linear regulators - System rail buck-or-boost (BoB) SMPS regulator - RGB LED drivers and camera flash drivers - Overtemperature protection - SPMI 9 GPIOs 143-pin fanout wafer-level nanoscale package (FOWNSP143)

Subsystem / Connectors	Feature Set	Description	Specification
		PM8350B	The PM8350B/PM8350BH/PM8350BHS interface PMIC, supplements the master core PM8350 PMIC and the slave core PM8350C PMIC to integrate all wireless handset power management, general housekeeping, and user interface support functions into an integrated three-IC solution. ▪ Best in class high-efficiency 6A 3-level buck/Div2 charge pump combination architecture Li-ion battery charger (> 97% peak efficiency) ▪ Dedicated paths for USB and wireless charging with charger operation up to 21.5 V (25 V absolute max) ▪ Supports up to a 6A charge current (single path) and a 12 A charge current (dual path using SMB1396/SMB1398 companion ICs) ▪ Supports USB Type-C Specification Rev. 1.4 and USB Power Delivery Specification Rev. 3.0 ▪ Supports Qualcomm® Quick Charge™ 2.0, Quick Charge 3.0, and Quick Charge 4.0 technology ▪ Supports 2S battery with charge balancing (SMB1399 needed). See Section 1.3 for details. ▪ Simultaneous wireless charging and OTG support using external 5 V boost ▪ Simultaneous USB charging and reverse wireless charging support using external 5 V boost ▪ Qualcomm battery gauge with improved SoC accuracy ▪ Analog-to-digital converter (ADC) subsystem with analog multiplexer (AMUX) and voltage/current analog-to-digital converter (VADC/IADC) ▪ Battery interface module (BIM) that includes battery serial interface (BSI) or PMIC serial interface (PSI) and battery missing detection (BMD) ▪ Battery current limiting (BCL) subsystem that provides both hardware and software alarms, which are used by the system for reducing battery current. ▪ Triple supply high performance and high efficiency boost regulators for OLED panels ▪ Full H-bridge haptics driver with dedicated 5 V to 10 V programmable boost for LRA application and SoundWire interface. See Section 1.3 for details. ▪ SPMI interface ▪ Eight GPIOs ▪ 165-pin fan-out wafer-level nanoscale package (165 FOWNSP)

Subsystem / Connectors	Feature Set	Description	Specification
		PM8010	The PM8010 device is a dedicated camera PMIC that integrates all the necessary power management, housekeeping, and interface support functions into a single IC. ▪ I2C/SPMI interface support for interrupt communication (default slave I2C address is 0x08). ▪ Seven low-dropout linear (LDO) linear regulators ▪ Overtemperature protection ▪ I2C/SPMI with 400 kHz and 2.2 kΩ pull-up resistor ▪ Undervoltage lockout (UVLO) support ▪ High PSRR and low-noise regulators for camera sensor applications ▪ 20-pin wafer-level picoscale package (WLPSP)
		PMR735A	The PMR735A device is a power management IC that is designed to power RF circuits. It integrates all the necessary power management, housekeeping, and interface support functions into a single IC. ▪ One HFS515 SMPS and two FTS520 SMPS ▪ Seven LDO linear regulators ▪ On-chip ADC ▪ Four GPIOs ▪ SPMI ▪ 48-pin wafer-level nanoscale package (WLNSP)
Memory	LPDDR5	12 GB LPDDR5	LPDDR5 up to 3200MHz
	UFS	256 GB UFS3.0	UFS 3.0
Connectivity	Wi-Fi 2.4 GHz/5GHz via WCN6856 – PCIE,UART	Wi-Fi WCN6856 Wi-Fi + BT Combo Chip	802.11a/b/g/n/ac/ax 2.4/5.0 GHz
	BT 2.4 GHz via WCN6856 – UART / SLIMbus	Wi-Fi WCN6856 Wi-Fi + BT +FM Combo Chip	Support BT 5.2 + HS and backward compatible with BT 1.x, 2.x + EDR
Camera Expansion	4 x 4-Lane MIPI CSI DPHY or 4 x 3-Trio MIPI CSI CPHY	1 x 160-pin Camera Connector for CSI0, CSI1, CSI2 and CSI3	D-PHY v1.2 2.5 Gbps per lane C-PHY v1.2 10.26 Gbps per T
	1 x 4-Lane/1 x 2-Lane MIPI CSI DPHY or 4 x 3-Trio MIPI CSI CPHY	1 x 60-pin Camera Connector for CSI4, CSI5 on expansion card	D-PHY v1.2 2.5 Gbps per lane C-PHY v1.2 10.26 Gbps per T
Sensor Expansion	2 x 60-pin GenX connector	60-pin GenX connector	I3C, I2C, SPI

Subsystem / Connectors	Feature Set	Description	Specification
	1 x 30-pin Vertigo connector	30-pin Vertigo connector	I3C, I2C, SPI
	1 x 60-pin GenX connector	60-pin GenX connector on expansion card	I3C, I2C, SPI
	1 x 24-pin legacy sensor connector	24-pin legacy sensor connector on expansion card	I3C/I2C, SPI
Audio Onboard	Audio Codec WCD9385	1x Headset headphone output	Analog differential output
		1x loud-speaker output	Digital output
		1x Earpiece output	Analog differential output
		4 x Digital MIC input	Digital input
		2 x Analog MIC input(reserved interface on expansion card)	Analog differential input
		1 x I2S (reserved interface on expansion card)	Digital input/output
Display Expansion	2 x 4-Lane MIPI DSI DPHY or 2 x 3-Trio MIPI DSI CPHY	2 x 60-pin High Speed Display Connector	D-PHY1.2 2.5 Gbps per lane C-PHY1.1 2.5 Gsymbols per trio
HDMI	Resolution Up to 4K 60Hz via LT9611UXC	2-Port MIPI to HDMI2.0 Converter	Compliant with HDMI2.0

3.3 Dip switch configuration option

There are two DIP switches S5701 and S5702 on the bottom side of the HDK8450 Main board. The 8-bit switch of S5701 allows the user to control the system configuration and boot option.

Table 3-3 Dip Switch S5701 Configuration

Function	DIP Switch	Description	Notes
FORCED_USB_BOOT	S5701-1	Enable Emergency Download when DIP switch turned on. Controlled by GPIO47.	Default out of the box configuration is OFF
FASTBOOT(0)	S5701-2	Enable other boot configuration when DIP switch turned on. Controlled by GPIO67.	Default out of the box configuration is OFF.

Function	DIP Switch	Description	Notes
FASTBOOT(1)	S5701-3	Enable other boot configuration when DIP switch turned on. Controlled by GPIO158.	Default out of the box configuration is OFF
FASTBOOT(2)	S5701-4	Enable other boot configuration when DIP switch turned on. Controlled by GPIO161.	Default out of the box configuration is OFF
WDG_DISABLE	S5701-5	Enable WATCHDOG_DISABLE when DIP switch turned on. Controlled by GPIO164.	Default out of the box configuration is OFF.
BUCK_PWR_CTRL	S5701-6	Manually disable external 5V/3.3V buck output.	Default out of the box configuration is OFF
M2_PWR_CTRL	S5701-7	Manually disable 1.8V and 3.3V power supply	Default out of the box configuration is OFF
AOC_DET	S5701-8	Manually connect USB-C headset detect signal to CODEC Default is disconnected.	Default out of the box configuration is OFF.

The detailed configuration for boot sequence choice is listed in below table.

Table 3-4 Boot Sequence

FAST_BOOT	GPIO#				Boot Sequence
	59	161	158	67	
0b0000	0	0	0	0	Default: UFS0 -> SDC2 -> eDL(USB0)
0b0001	0	0	0	1	SDC2 -> UFS0 -> eDL(USB0)
0b0010	0	0	1	0	SDC2 -> eDL (USB0)
0b0011	0	0	1	1	USB0 boot only
0b0100	0	1	0	0	QSPI -> SDC2 -> eDL (USB0)
0b0101	0	1	0	1	SPI -> SDC2 -> eDL (USB0)
0b0110	0	1	1	0	UFS0 -> eDL (USB0)
0b0111	0	1	1	1	Same as Fastboot[3:0] = 0000 UFS0 -> SDC2 -> eDL(USB0)

The 8-bit switch of S5702 allows the user to do some general configuration.

Table 3-5 Dip Switch S5702 Configuration

Function	DIP Switch	Description	Notes
SYS_PWR_CTRL	S5702-1	Manually disable external 3.9V buck output. Default the buck is ON when DC adapter is plugged in.	Default out of the box configuration is OFF.
BATT_FET_CTRL	S5702-2	Manually disable battery supply path to system. Default the path is controlled by hardware automatically.	Default out of the box configuration is OFF.
BATT_ID0	S5702-3	Connect BATT_ID to resistor 1(7.5K). Default is disconnected.	Default out of the box configuration is OFF

Function	DIP Switch	Description	Notes
BATT_ID1	S5702-4	Connect BATT_ID to resistor 2(reserved). Default is disconnected.	Default out of the box configuration is OFF
BATT_THERM0	S5702-5	Connect BATT_THERM to resistor 1(100K). Default is disconnected.	Default out of the box configuration is OFF
BATT_THERM1	S5702-6	Connect BATT_THERM to resistor 1(reserved). Default is disconnected.	Default out of the box configuration is OFF
DSI0_SW	S5702-7	Manually select DSI0 data path between display panel and HDMI. Default route DSI0 to the display panel.	Default out of the box configuration is OFF
DSI1_SW	S5702-8	Manually select DSI1 data path between display panel and HDMI. Default route DSI1 to the display panel.	Default out of the box configuration is OFF

3.3.1 Development kit setup – display mode(default)

Display panel is the default output. Make sure switch7&8 of S5702 is **OFF** before turning on the device.

Software detects the status of Switch7&8 and automatically configure the output to support display panel.

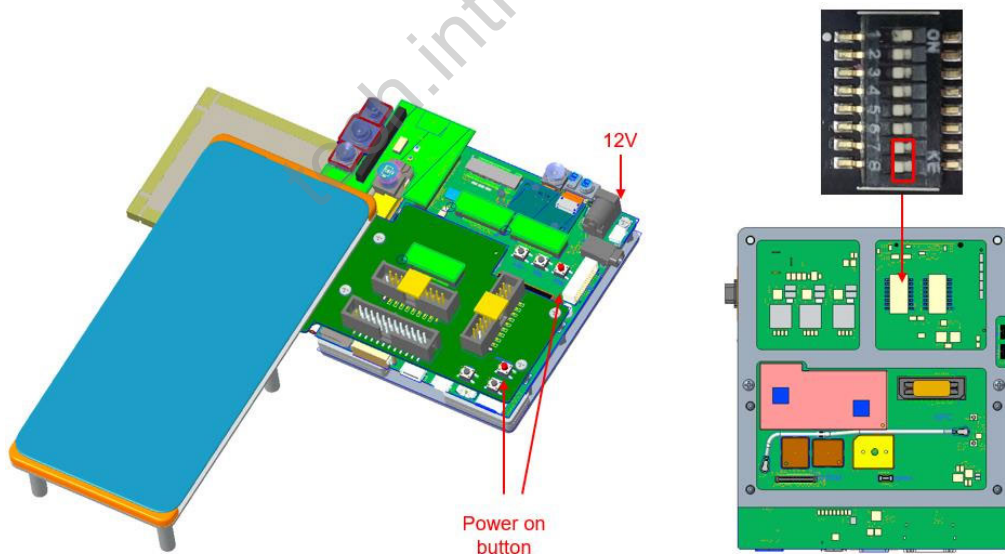


Figure 3-3 Display Mode Setup

3.3.2 Development kit setup – HDMI mode

Make sure switch7&8 of S5702 is **ON** before turning on the device.

Software detects the status of Switch8 and automatically config the output to support HDMI output.

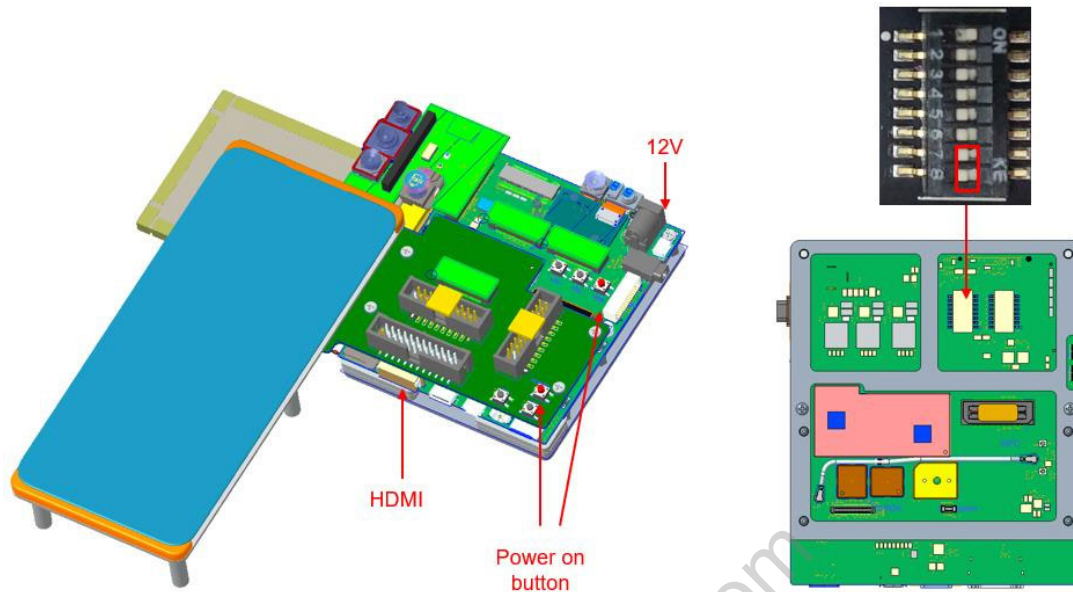


Figure 3-4 HDMI Mode Setup

3.3.3 Development kit setup – emergency download

Put switch1 of S5701 on when entering Emergency Download (EDL) mode. Remember to switch it back when exiting EDL mode and entering normal power on mode.

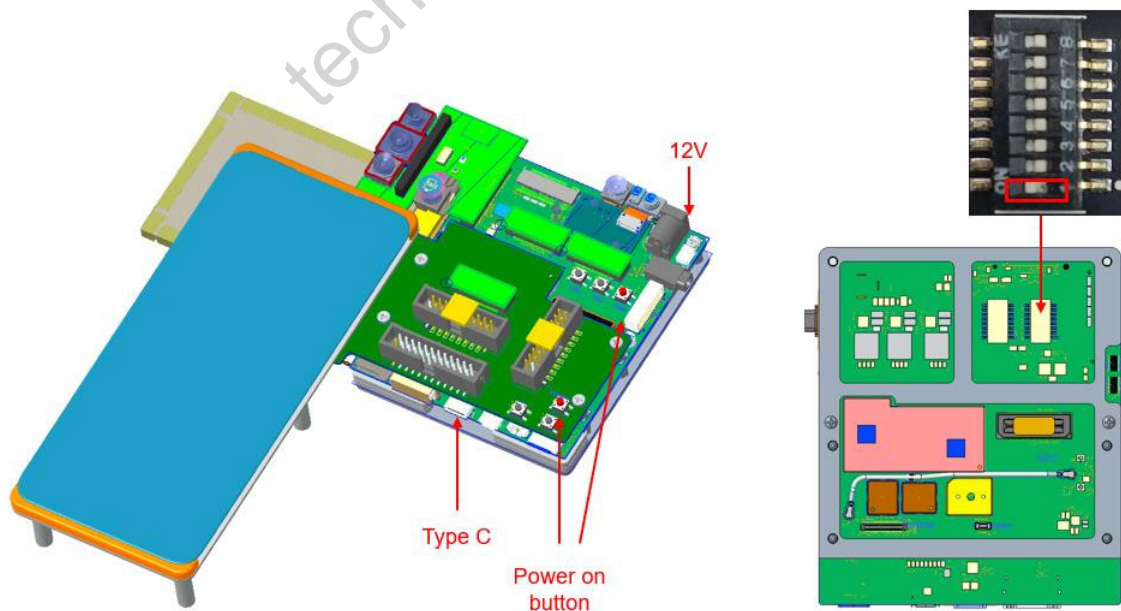


Figure 3-5 Emergency Download Setup

3.4 Main board connectors and features

Table below lists the connectors, expansions and their usages on the main board.

Table 3-6 Main Board Connectors and Features

Domain	Description	Specification	Usage
Power	AC / Barrel charger	12V/5A DC Power Supply	Power Supply
	Battery connector	2 x 16-pin connector	For providing power from 4.4V/3780mAh battery(optional)
Debug Serial via USB	Debug Serial UART console over USB for development	USB Micro B connector	Development Serial Connector for debug output via USB
Buttons	Power On	SMD Button	Red Power Button for Suspend / Resume and Power off
	Volume +	SMD Button	Volume + Key
	Volume –	SMD Button	Volume – Key
Headset Jack	Supported by WCD9385	Normal Open, support US Standard CTIA headset by default	Audio headset support
USB	1x Type-C USB3.1	Type-C USB3.1 GEN2 10Gbps	Data transfer
HDMI	HDMI2.0	4K60Hz	HDMI output
SD Card	Micro SD card	4bit Micro SD card	External Storage
Camera Expansion	1 x 160-pin connector with MCLK, GPIOs, CCI	Supports CSI0, CSI1, CSI2 and CSI3 via the 160-pin connector 4x 4-Lane MIPI-CSI DPHY1.2 or 4x 3-Trio MIPI CSI CPHY1.0 - External flash driver control - Support for 3D camera configuration - Separate MCLK / CCI control	- Single Rear, Front camera - Dual Camera - Iris Camera
Sensor Expansion	2x 60 pin-GenX sensor connector	60-pin GenX sensor connector	Connector for sensor card
Vertigo	CDP sensor connector	CDP sensor connector	Connection for sensor bar
High Speed Expansion	60-pin B2B connector for DSI0 and CSI4/5 signals. Compliant with 96boards HS expansion spec	Supports DSI0 via the 60-pin connector 4-Lane MIPI-CSI DPHY1.2 or 3-Trio MIPI CSI CPHY1.1 - I2C/SPI/GPIO	FHD 1080x2340 120Hz panel is supported on HDK8450

Domain	Description	Specification	Usage
High Speed Expansion-2	60-pin B2B connector for DSI1, touch, display powers(OLED bias, display, touch) to supplement the display support. JTAG, MICs, sensor powers are also included.	Supports DSI1 via the 60-pin connector 4-Lane MIPI-CSI DPHY1.2 or 3-Trio MIPI CSI CPHY1.1 - I2C/SPI/GPIO Power suppliers for OLED and sensors Support JTAG signals Support MIC input 2 Analog MICs - MIC bias supplier	The supplement to support dual DSI display panel with power suppliers. Also connect JTAG, MICs to expansion card
Low Speed Expansion	1x 40-pin B2B connector for low speed signals Compliant with 96boards HS expansion spec	Supports SPI/I3C/I2C/UART/I2S/ and GPIOs via the 40-pin connector	Support sensor connectors and Power/Volume Buttons on the Expansion Card
WLAN Antenna	2x PCB Antenna	2.4GHz 5GHz	On-board antenna
LED	1x Power LED	Blue	Adaptor in
	4x User LED	- Red: PMIC Driven - Green: PMIC Driven - Blue: PMIC Driven - Green2: GPIO Control	User LED
	1x WLAN LED (reserve) 1x BT LED (reserve)	- Orange: GPIO Control - Blue: GPIO Control	WLAN on BT on

The following sections provide in depth information on each expansion connectors on the Main board. The information listed below is of particular use for those who want to interface other external hardware devices with HDK8450. Before connecting anything to the development kit, ensure the device meets the specific hardware requirements of the processor.

3.4.1 DC power jack

The HDK development kit is powered by a 12V DC adaptor through J3001. Starting from the power jack, the 12V power supply branches off into different voltage rails via step down converters. The system is powered by 3.9V via Silergy step down converter U5101.

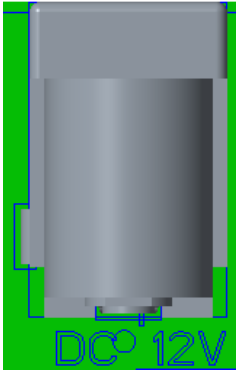


Figure 3-6 J5101 12V DC Power Jack

The Main Board consists of lots of PMIC modules, including PMK8450, PM8450, PM8350, PM8350C, PM8350B, PM8010 and PMR735A. The functionalities of the PMICs are outlined below.

- Source various regulated power rails
- Source system clock
- Support for battery charging.

The charging port in PM8350B is configurable on the platform. The main board can either be powered by 12V power adaptor or battery. 12V adaptor is the default power source for HDK8450.

3.4.2 Battery connector

Battery is an optional power source for HDK8450, system will automatically select 12V DC-IN as default power supply when both of them are available.

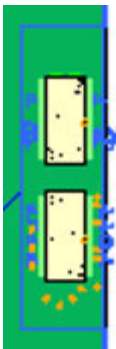


Figure 3-7 J2401 and J2402 Battery Connector

3.4.3 Debug serial UART over USB

The UART connection used on the HDK8450 is a USB micro B connector. This debug UART is available over USB via the FTDI FT232RQ chip on the Main board. To get the serial terminal working with a PC, user needs to ensure that the appropriate FTDI drivers are installed.

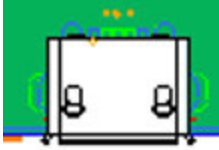


Figure 3-8 J5301 Debug UART over USB

3.4.4 Headset jack

The headset jack is standard 3.5mm , Normal Open style. The hardware supports US Standard CTIA headset by default.



Figure 3-9 J4501 Headset Jack

3.4.5 USB3.1 Type-C connector

The on board Type-C connector supports USB3.1 Gen2 10Gbps. It also supports DisplayPort1.4.

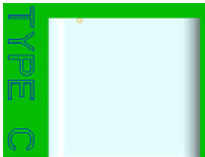


Figure 3-10 J3701 USB3.1 Type-C Connector

3.4.6 HDMI connector

The HDMI type A connector enables the HDK8450 development platform to connect to an external HDMI monitor/ television via an HDMI cable. The LT9611UXC MIPI DSI to HDMI2.0 bridge IC features a resolution up to 4K60Hz. The default output is 3980x2060 which is SW configurable.

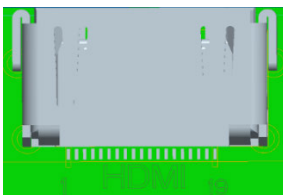


Figure 3-11 J5001 HDMI Connector

3.4.7 SD socket

The on-board Micro SD socket supports micro SD card speed up to UHS104 (SDR104).

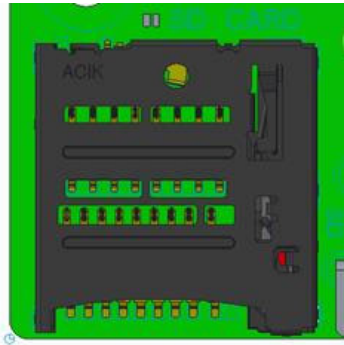


Figure 3-12 J5501 SD socket

3.4.8 Rear Camera card connector

The HDK8450 development kit supports four 4-Lane MIPI camera interfaces via a 160pin connector.

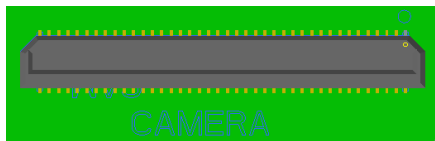


Figure 3-13 J4701 Rear Camera Connector

The following are some features of the camera connectors:

- 4 x 4-Lane MIPI CSI signals, CSI0, CSI1, CSI2 and CSI3
- Support for 3D camera configuration
- Separate I2C control (CCI0/1/2/3)
- On-board Amphenol 11828-1DA connector for expanding MIPI, MCLK, CCI, GPIOs and Power rails.
- The recommended mating connector on a camera card is Amphenol 11826-BDA.
- If existing 30-12807-1 camera card is used, it will be on the same side with display, customer could adjust the angle in software if needed

Table below lists the pinout of the camera connector.

Table 3-7 Camera Expansion Connector Pinout

Pin#	Signal	Description	Pin#	Signal	Description
A1	GND		B1	GND	
A2	CSI0_C2_LN3_M	CSI0 data	B2	CSI2_C2_LN3_M	CSI2 data
A3	CSI0_B2_LN3_P	CSI0 data	B3	CSI2_B2_LN3_P	CSI2 data
A4	CSI0_A2_LN2_M	CSI0 data	B4	CSI2_A2_LN2_M	CSI2 data
A5	CSI0_C1_LN2_P	CSI0 data	B5	CSI2_C1_LN2_P	CSI2 data

Pin#	Signal	Description	Pin#	Signal	Description
A6	CSI0_B1_LN1_M	CSI0 data	B6	CSI2_B1_LN1_M	CSI2 data
A7	CSI0_A1_LN1_P	CSI0 data	B7	CSI2_A1_LN1_P	CSI2 data
A8	CSI0_C0_LN0_M	CSI0 data	B8	CSI2_C0_LN0_M	CSI2 data
A9	CSI0_B0_LN0_P	CSI0 data	B9	CSI2_B0_LN0_P	CSI2 data
A10	CSI0_A0_CLK_M	CSI0 clock	B10	CSI2_A0_CLK_M	CSI2 clock
A11	CSI0_NC_CLK_P	CSI0 clock	B11	CSI2_NC_CLK_P	CSI2 clock
A12	GND		B12	GND	
A13	CSI1_C2_LN3_M	CSI1 data	B13	CSI3_C2_LN3_M	CSI3 data
A14	CSI1_B2_LN3_P	CSI1 data	B14	CSI3_B2_LN3_P	CSI3 data
A15	CSI1_A2_LN2_M	CSI1 data	B15	CSI3_A2_LN2_M	CSI3 data
A16	CSI1_C1_LN2_P	CSI1 data	B16	CSI3_C1_LN2_P	CSI3 data
A17	CSI1_B1_LN1_M	CSI1 data	B17	CSI3_B1_LN1_M	CSI3 data
A18	CSI1_A1_LN1_P	CSI1 data	B18	CSI3_A1_LN1_P	CSI3 data
A19	CSI1_C0_LN0_M	CSI1 data	B19	CSI3_C0_LN0_M	CSI3 data
A20	CSI1_B0_LN0_P	CSI1 data	B20	CSI3_B0_LN0_P	CSI3 data
A21	CSI1_A0_CLK_M	CSI1 clock	B21	CSI3_A0_CLK_M	CSI3 clock
A22	CSI1_NC_CLK_P	CSI1 clock	B22	CSI3_NC_CLK_P	CSI3 clock
A23	GND	Ground	B23	GND	Ground
A24	CAM_I3C_SDA	I3C data	B24	VREG_L4J_1P8	1.8V power supplier
A25	CAM_I3C_SCL	I3C clock	B25	VREG_L6I_2P8	2.8V power supplier
A26	GND	Ground	B26	GND	Ground
A27	REAR_DMIC_DAT_A	I2C data	B27	FLK_ALS_SPI_MISO	SPI MISO
A28	REAR_DMIC_CLK	I2C clock	B28	FLK_AS_L_SPI_CLK	SPI_CLK
A29	GND	Ground	B29	VGRE_S10B_1P8	1.8V power supplier
A30	VREG_L4I_2P8	2.8V power supplier	B30	GND	Ground
A31	GND	Ground	B31	VREG_L3I_2P9	2.9V power supplier
A32	VREG_L3J_1P8	1.8V power supplier	B32	FLASH_STROBE1	Flash strobe
A33	GND	Ground	B33	GND	Ground
A34	VPH_PWR	3.9V power supply	B34	VPH_PWR	3.9V power supply
A35	GND	Ground	B35	GND	Ground
A36	VREG_L1I_1P1	1.1V power supplier	B36	VREG_L1I_1P1	1.1V power supplier
A37	GND	Ground	B37	GND	Ground
A38	VREG_S12B_1P25	1.25V power supplier	B38	VREG_S12B_1P25	1.25V power supplier
A39	GND	Ground	B39	GND	Ground
A40	VREG_L7I_2P96	2.96V power supplier	B40	VREG_L7I_2P96	2.96V power supplier

Pin #	Signal	Description	Pin#	Signal	Description
C1	GND	Ground	D1	CCI_I2C0_SDA	CCI
C2	GND	Ground	D2	CCI_I2C0_SCL	CCI
C3	GND	Ground	D3	GND	Ground
C4	GND	Ground	D4	CCI_I2C1_SDA	CCI
C5	GND	Ground	D5	CCI_I2C1_SCL	CCI
C6	GND	Ground	D6	GND	Ground
C7	GND	Ground	D7	CCI_I2C2_SDA	CCI
C8	GND	Ground	D8	CCI_I2C2_SCL	CCI
C9	GND	Ground	D9	GND	Ground
C10	GND	Ground	D10	LRF_IRQ	GPIO44
C11	GND	Ground	D11	NC	
C12	GND	Ground	D12	GND	Ground
C13	GND	Ground	D13	CAM_MCLK0	GPIO100
C14	GND	Ground	D14	CAM0_RESET_N	GPIO17
C15	GND	Ground	D15	GND	Ground
C16	GND	Ground	D16	CAM_MCLK1	GPIO101
C17	GND	Ground	D17	CAM1_RESET_N	GPIO16
C18	GND	Ground	D18	GND	
C19	GND	Ground	D19	CAM_MCLK2	GPIO102
C20	GND	Ground	D20	CAM2_RESET_N	GPIO106
C21	GND	Ground	D21	GND	Ground
C22	GND	Ground	D22	CAM_MCLK3	GPIO103
C23	CAM_SPI0_CLK_CONN	GPIO54	D23	CAM3_RESET_N	GPIO117
C24	CAM_SPI0_MOSI_CONN	GPIO53	D24	CAM_FLASH_THERM	PM8350 AMUX2
C25	CAM_SPI0_MISO_CONN	GPIO52	D25	REAR_TOF_THERM	PM8350 GPIO1
C26	CAM_SPI0_CS_N_CONN	GPIO55	D26	CSI_MUX_EN	GPIO60
C27	VREG_L2I_1P05	1.05V power supplier	D27	CSI_MUX_SEL	GPIO61
C28	CAM_SPI1_CLK	GPIO74	D28	LRF_IRQ	GPIO66
C29	CAM_SPI1_MOSI	GPIO73	D29	WIDE_RFC_THERM	PM8350 AMUX4
C30	CAM_SPI1_MISO	GPIO72	D30	APPS_I2C_SDA	I2C data
C31	CAM_SPI1_CS_N	GPIO75	D31	APPS_I2C_SCL	I2C clock
C32	GND	Ground	D32	VREG_L6J_2P8	2.8 power supplier
C33	VREG_L7J_3P3	3.3V power supplier	D33	WCD_MIC_BIAS1	MIC bias
C34	GND	Ground	D34	GND	Ground
C35	VREG_TOF_BOB	TOF power supplier	D35	VREG_TOF_BOB	TOF power supplier
C36	GND	Ground	D36	GND	Ground
C37	CAM_FLASH_2_3	Flash LED2	D37	CAM_FLASH_2_3	Flash LED2

Pin #	Signal	Description	Pin#	Signal	Description
C38	GND	Ground	D38	GND	Ground
C39	CAM_FLASH_1_4	Flash LED1	D39	CAM_FLASH_1_4	Flash LED1
C40	GND	Ground	D40	GND	Ground

3.4.9 Front camera card connector



Figure 3-6 J4601 Front Camera Card Connector

Table 3-7 Front Camera Card Connector Pinout

Pin#	Signal	Description	Pin#	Signal	Description
J901[1]	VREG_L2I_1P05	1.05V power supplier	J901[2]	VREG_L3J_1P8	1.8V power supplier
J901[3]	VREG_L2I_1P05	1.05V power supplier	J901[4]	GND	Ground
J901[5]	VREG_L2I_1P05	1.05V power supplier	J901[6]	CCI_I2C2_SDA	GPIO114
J901[7]	GND	Ground	J901[8]	CCI_I2C2_SSC	GPIO115
J901[9]	CSI5_C2_LN3_M	CSI5 Data	J901[10]	GND	Ground
J901[11]	CSI5_C2_LN3_P	CSI5 Data	J901[12]	CSI5_A2_LN2_M	CSI5 Data
J901[13]	GND	Ground	J901[14]	CSI5_C1_LN2_P	CSI5 Data
J901[15]	CSI5_B1_LN1_M	CSI5 Data	J901[16]	GND	Ground
J901[17]	CSI5_B1_LN1_P	CSI5 Data	J901[18]	CSI5_C0_LN0_M	CSI5 Data
J901[19]	GND	Ground	J901[20]	CSI5_B0_LN0_P	CSI5 Data
J901[21]	CSI5_A0_CLK_M	CSI5 Clock	J901[22]	GND	Ground
J901[23]	CSI5_A0_CLK_P	CSI5 Clock	J901[24]	CCI_I2C3_SDA	GPIO208
J901[25]	GND	Ground	J901[26]	CCI_I2C3_SCL	GPIO209
J901[27]	CAM4_RESET_N	GPIO119	J901[28]	GND	Ground
J901[29]	VREG_TOF_BOB	3.6V power supplier	J901[30]	VREG_TOF_BOB	3.6V power supplier
J901[31]	VREG_TOF_BOB	3.6V power supplier	J901[32]	VREG_TOF_BOB	3.6V power supplier
J901[33]	CAM5_RESET_N	GPIO118	J901[34]	CAM_MCLK6	GPIO106
J901[35]	CAM_MCLK5	GPIO105	J901[36]	VREG_L5J_2P8	2.8V power supplier
J901[37]	GND	Ground	J901[38]	VREG_L2J_1P05	1.05V power supplier
J901[39]	CSI4_C2_LN3_M	CSI4 Data	J901[40]	CAM_MCLK4	GPIO104

Pin#	Signal	Description	Pin#	Signal	Description
J901[41]	CSI4_B2_LN3_P	CSI4 Data	J901[42]	GND	Ground
J901[43]	GND	Ground	J901[44]	CSI4_A2_LN2_M	CSI4 Data
J901[45]	CSI4_B1_LN1_M	CSI4 Data	J901[46]	CSI4_C1_LN2_P	CSI4 Data
J901[47]	CSI4_A1_LN1_P	CSI4 Data	J901[48]	GND	Ground
J901[49]	GND	Ground	J901[50]	CSI4_C0_LN0_M	CSI4 Data
J901[51]	CSI4_A0_CLK_M	CSI4 Clock	J901[52]	CSI4_B0_LN0_P	CSI4 Data
J901[53]	CSI4_NC_CLK_P	CSI4 Clock	J901[54]	GND	Ground
J901[55]	GND	Ground	J901[56]	AON_DRI_INT	GPIO85
J901[57]	VREG_L7I_2P96	2.96V power supplier	J901[58]	GND	Ground
J901[59]	AON_RESET_N	GPIO108	J901[60]	VREG_L6I_2P8	2.8V power supplier

3.4.10 Sensor1 expansion GenX connector

The sensor expansion connector allows for a 60-pin connection to an external sensor card. There are two GenX connectors on the Main Board. For ordinary usage, use Sensor1.

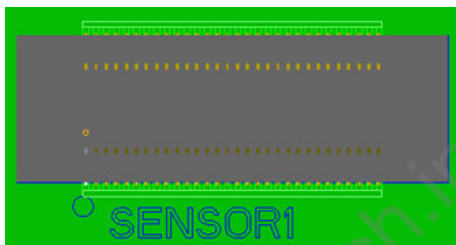


Figure 3-14 J5902 Sensor1 Expansion GenX Connector

Table 3-8 Sensor1 Expansion GenX Connector Pinout

Pin#	Signal	Description	Pin#	Signal	Description
J5902[1]	NC		J5902[2]	NC	
J5902[3]	NC		J5902[4]	IMU_SPI_MOSI	GPIO 193
J5902[5]	NC		J5902[6]	IMU_SPI_MISO	GPIO 192
J5902[7]	NC		J5902[8]	IMU_SPI_SCLK	GPIO 194
J5902[9]	NC		J5902[10]	IMU_SPI_CS_N	GPIO 195
J5902[11]	NC		J5902[12]	NC	
J5902[13]	NC		J5902[14]	NC	
J5902[15]	NC		J5902[16]	NC	
J5902[17]	NC		J5902[18]	SAR_INT_N	GPIO63
J5902[19]	NC		J5902[20]	ALSP_INT_N	GPIO14
J5902[21]	NC		J5902[22]	HALL_INT_N	GPIO13
J5902[23]	VREG_L8C_1P8_SNS	1.8V sensor digital power	J5902[24]	NFC_INT_REQ	GPIO46
J5902[25]	NC		J5902[26]	IMU2_INT_1P2	GPIO46
J5902[27]	NC		J5902[28]	PRESS_INT	GPIO83

Pin#	Signal	Description	Pin#	Signal	Description
J5902[29]	NC		J5902[30]	IMU1_INT_1P2	GPIO15
J5902[31]	NC		J5902[32]	MAG_INT_N_1P2	GPIO62
J5902[33]	NC		J5902[34]	SENSOR_I3C_SCL	GPIO 188
J5902[35]	NC		J5902[36]	SENSOR_I3C_SDA	GPIO 189
J5902[37]	NC		J5902[38]	VREG_L7C_3P0_S NS	3.0V sensor analog power
J5902[39]	NC		J5902[40]	SENSOR_SPI2_MO SI	GPIO 199
J5902[41]	SENSOR_SPI2_ MISO	GPIO198	J5902[42]	SENSOR_SPI2_CLK	GPIO 200
J5902[43]	SENSOR_SPI2_ CS_N	GPIO201	J5902[44]	NC	
J5902[45]	MAG_I2C_SDA_ 1P2	GPIO190	J5902[46]	NC	
J5902[47]	NC		J5902[48]	NC	
J5902[49]	NC		J5902[50]	NC	
J5902[51]	NC		J5902[52]	SENSOR_I2C_SDA	GPIO 196
J5902[53]	NC		J5902[54]	SENSOR_I2C_SCL	GPIO 197
J5902[55]	NC		J5902[56]	NC	
J5902[57]	NC		J5902[58]	NC	
J5902[59]	NC		J5902[60]	NC	

3.4.11 Sensor2 expansion GenX connector

For dual SPI and I2C application, Sensor2 can be used with proper software setting.

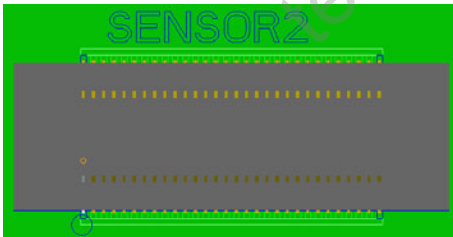


Figure 3-15 J5903 Sensor2 Expansion GenX Connector

Table 3-9 Sensor2 Expansion GenX Connector Pinout

Pin#	Signal	Description	Pin#	Signal	Description
J5903[1]	NC		J5903[2]	NC	
J5903[3]	NC		J5903[4]	SENSOR_SPI2_MO SI	GPIO199
J5903[5]	NC		J5903[6]	SENSOR_SPI2_MI SO	GPIO198
J5903[7]	NC		J5903[8]	SENSOR_SPI2_CL K	GPIO200
J5903[9]	NC		J5903[10]	SENSOR_SPI2_CS _N	GPIO201

Pin#	Signal	Description	Pin#	Signal	Description
J5903[11]	NC		J5903[12]	NC	
J5903[13]	NC		J5903[14]	NC	
J5903[15]	NC		J5903[16]	NC	
J5903[17]	NC		J5903[18]	NFC_INT_REQ	GPIO46
J5903[19]	NC		J5903[20]	MAG_INT_N	GPIO119
J5903[21]	NC		J5903[22]	IMU2_INT	GPIO12
J5903[23]	VREG_L8C_1P8_SNS	1.8V sensor digital power	J5903[24]	SAR_INT_N	GPIO63
J5903[25]	NC		J5903[26]	PRESS_INT	GPIO83
J5903[27]	NC		J5903[28]	IMU1_INT	GPIO15
J5903[29]	NC		J5903[30]	ALSP_INT_N	GPIO14
J5903[31]	NC		J5903[32]	HALL_INT_N	GPIO13
J5903[33]	NC		J5903[34]	MAG_I2C_SCL	GPIO191
J5903[35]	NC		J5903[36]	MAG_I2C_SDA	GPIO190
J5903[37]	NC		J5903[38]	VREG_L7C_3P0_SNS	3.0V sensor analog power
J5903[39]	NC		J5903[40]	IMU_SPI_MOSI	GPIO193
J5903[41]	IMU_SPI_MISO	GPIO192	J5903[42]	IMU_SPI_SCLK	GPIO194
J5903[43]	IMU_SPI_CS_N	GPIO195	J5903[44]	NC	
J5903[45]	NC		J5903[46]	NC	
J5903[47]	NC		J5903[48]	NC	
J5903[49]	NC		J5903[50]	NC	
J5903[51]	NC		J5903[52]	MAG_I2C_SDA	GPIO196
J5903[53]	NC		J5903[54]	MAG_I2C_SCL	GPIO197
J5903[55]	NC		J5903[56]	NC	
J5903[57]	NC		J5903[58]	NC	

Pin#	Signal	Description	Pin#	Signal	Description
J5903[59]	NC		J5903[60]	NC	

3.4.12 Vertigo sensor connector

The HDK8450 has a 30-pin connector (449-53935-0030) to support Vertigo13 or following sensor cards that are supported on CDP8350.

Vertigo12 sensor card is designed and managed by the sensor core team.

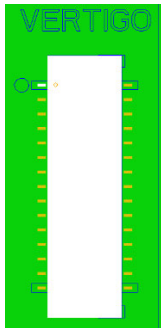


Figure 3-16 J5901 Vertigo Connector

3.4.13 High speed expansion connector

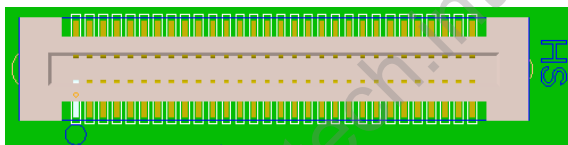


Figure 3-17 J6001 High Speed Expansion Connector

High speed expansion connector provides DSI0, CSI4/5 and other signals such as SPI/USB HS. This interface is compliant with 96boards spec.

Table 3-10 High Speed Expansion Connector Pinout

Pin#	Signal	Description	Pin#	Signal	Description
J6001[1]	SENSOR_SPI2_MOSI	GPIO199	J6001[2]	NC	
J6001[3]	NC		J6001[4]	NC	
J6001[5]	NC		J6001[6]	GND	
J6001[7]	SENSOR_SPI2_CS_N	GPIO201	J6001[8]	NC	
J6001[9]	SENSOR_SPI2_CLK	GPIO200	J6001[10]	NC	
J6001[11]	SENSOR_SPI2_MISO	GPIO198	J6001[12]	GND	
J6001[13]	GND		J6001[14]	NC	
J6001[15]	NC		J6001[16]	NC	

Pin#	Signal	Description	Pin#	Signal	Description
J6001[17]	NC		J6001[18]	GND	
J6001[19]	GND	Ground	J6001[20]	NC	
J6001[21]	MIPI_DSI0_CLK_P_DISP	DSI0 clock	J6001[22]	NC	
J6001[23]	MIPI_DSI0_CLK_M_DISP	DSI0 clock	J6001[24]	GND	
J6001[25]	GND	Ground	J6001[26]	NC	
J6001[27]	MIPI_DSI0_LANE0_P_DISP	DSI0 data	J6001[28]	NC	
J6001[29]	MIPI_DSI0_LANE0_M_DISP	DSI0 data	J6001[30]	GND	
J6001[31]	GND	Ground	J6001[32]	NC	
J6001[33]	MIPI_DSI0_LANE1_P_DISP	DSI0 data	J6001[34]	NC	
J6001[35]	MIPI_DSI0_LANE1_M_DISP	DSI0 data	J6001[36]	NC	
J6001[37]	GND	Ground	J6001[38]	NC	
J6001[39]	MIPI_DSI0_LANE2_P_DISP	DSI0 data	J6001[40]	GND	
J6001[41]	MIPI_DSI0_LANE2_M_DISP	DSI0 data	J6001[42]	NC	
J6001[43]	GND	Ground	J6001[44]	NC	
J6001[45]	MIPI_DSI0_LANE3_P_DISP	DSI0 data	J6001[46]	GND	
J6001[47]	MIPI_DSI0_LANE3_M_DISP	DSI0 data	J6001[48]	NC	
J6001[49]	GND	Ground	J6001[50]	NC	
J6001[51]	DSI0_NC_CLK1_P_CON		J6001[52]	GND	
J6001[53]	DSI0_NC_CLK1_M_CON		J6001[54]	NC	
J6001[55]	GND	Ground	J6001[56]	NC	
J6001[57]	NC		J6001[58]	GND	
J6001[59]	NC		J6001[60]	NC	

3.4.14 High speed expansion connector-2

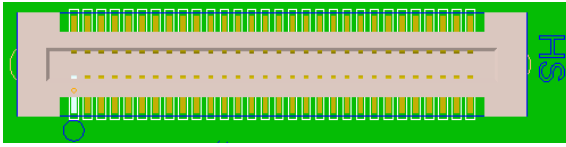


Figure 3-18 J6101 High Speed Expansion Connector -2

Table 3-11 High Speed Expansion Connector Pinout

Pin#	Signal	Description	Pin#	Signal	Description
J6101[1]	TS_SPI_CS_N	GPIO19	J6101[2]	VREG_ELVSS	OLED negative supplier
J6101[3]	TS_SPI_CLK	GPIO18	J6101[4]	VREG_ELVSS	OLED negative supplier
J6101[5]	TS_INT_N	GPIO21	J6101[6]	VREG_ELVDD	OLED positive supplier
J6101[7]	TS_RESET_N	GPIO20	J6101[8]	VREG_ELVDD	OLED positive supplier
J6101[9]	VDISP_SCTRL	PM8350 B	J6101[10]	VREG_OLEDB	OLED bias
J6101[11]	LCD_PWM	PM8350 C	J6101[12]	VREG_L12C_1P8	1.8V power supplier
J6101[13]	OLEDB_EN	PM8350 B	J6101[14]	VREG_L13C_3P0	3.0V power supplier
J6101[15]	DISP0_RESET_N	GPIO32	J6101[16]	VREG_L8C_1P8	1.8V power supplier
J6101[17]	MDP_VSYNC_P	GPIO86	J6101[18]	VREG_L3C_3P0	3.0V power supplier
J6101[19]	GND	Ground	J6101[20]	VREG_L8C_1P8_NS	1.8V power supplier
J6101[21]	MIPI_DSI1_CLK_P_DISP	DSI1 clock	J6101[22]	VREG_L7C_3P0_NS	3.0V power supplier
J6101[23]	MIPI_DSI1_CLK_M_DISP	DSI1 clock	J6101[24]	VREG_TP_VDDIO_1P8	1.8V power supplier
J6101[25]	GND	Ground	J6101[26]	VREG_DDIC	PM8350B
J6101[27]	MIPI_DSI1_LANE0_P_DISP	DSI1 data	J6101[28]	TS_INT2_N	
J6101[29]	MIPI_DSI1_LANE0_M_DISP	DSI1 data	J6101[30]	TP_SPI_MOSI_I2C_SCL	GPIO17
J6101[31]	GND	Ground	J6101[32]	TP_SPI_MISO_I2C_SDA	GPIO16
J6101[33]	MIPI_DSI1_LANE1_P_DISP	DSI1 data	J6101[34]	GND	Ground
J6101[35]	MIPI_DSI1_LANE1_M_DISP	DSI1 data	J6101[36]	WCD_MIC_BIAS3	MIC bias3
J6101[37]	GND	Ground	J6101[38]	WCD_MIC_BIAS1	MIC bias1
J6101[39]	MIPI_DSI1_LANE2_P_DISP	DSI1 data	J6101[40]	MIC01_DATA	Analog mic1

Pin#	Signal	Description	Pin#	Signal	Description
J6101[41]	MIPI_DSI1_LANE2_M_DISP	DSI1 data	J6101[42]	MIC01_CLK	Analog mic1
J6101[43]	GND	Ground	J6101[44]	I2C_MIC23_DATA	Analog mic5
J6101[45]	MIPI_DSI1_LANE3_P_DISP	DSI1 data	J6101[46]	I2C_MIC23_CLK	Analog mic5
J6101[47]	MIPI_DSI1_LANE3_M_DISP	DSI1 data	J6101[48]	GND	Ground
J6101[49]	GND	Ground	J6101[50]	SM_PS_HOLD	PS_HOLD
J6101[51]	EXP_JTAG_SRST_N	JTAG	J6101[52]	SNS2_SPI2_CS1_N_EXP	
J6101[53]	EXP_JTAG_TCK	JTAG	J6101[54]	EXP_JTAG_TRST_N	JTAG
J6101[55]	EXP_JTAG_TDI	JTAG	J6101[56]	VREG_S10B_1P8	1.8V power supplier
J6101[57]	EXP_JTAG_TDO	JTAG	J6101[58]	VPH_PWR	VPH_PWR
J6101[59]	EXP_JTAG_TMS	JTAG	J6101[60]	VPH_PWR	VPH_PWR

3.4.15 Low speed expansion connector

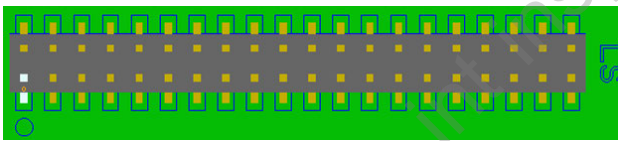


Figure 3-19 J6201 Low Speed Expansion Connector

Table 3-12 Low Speed Expansion Connector Pinout

Pin#	Signal	Description	Pin#	Signal	Description
J6201[1]	GND	Ground	J6201[2]	GND	ground
J6201[3]	EXP_UART_CTS	UART	J6201[4]	DEBUG_PMIC_PKD_N	PMK8350 KPD_PWR_N
J6201[5]	EXP_UART_TX	UART	J6201[6]	DEBUG_KEY_VOL_DO WN_N	PMK8350 RESET_N
J6201[7]	EXP_UART_RX	UART	J6201[8]	IMU_SPI_CLK	GPIO189
J6201[9]	EXP_UART_RFR	UART	J6201[10]	IMU_SPI_MISO	GPIO188
J6201[11]	NC		J6201[12]	IMU_SPI_CS_N	GPIO190
J6201[13]	NC		J6201[14]	IMU_SPI_MOSI	GPIO188
J6201[15]	SENSOR_I3C_SCL	GPIO184	J6201[16]	MI2S2_I2S_WS	GPIO122
J6201[17]	SENSOR_I3C_SDA	GPIO183	J6201[18]	MI2S2_I2S_SCK	GPIO120
J6201[19]	SENSOR_I2C_SCL	GPIO193	J6201[20]	MI2S2_I2S_DAT0	GPIO121
J6201[21]	SENSOR_I2C_SDA	GPIO194	J6201[22]	MI2S2_I2S_DAT1	GPIO124
J6201[23]	IMU1_INT	GPIO118	J6201[24]	DEBUG_KEY_VOL_UP_N	PM8350 GPIO06
J6201[25]	IMU2_INT	GPIO64	J6201[26]	MAG_INT_N	GPIO119

Pin#	Signal	Description	Pin#	Signal	Description
J6201[27]	ALSP_INT_N	GPIO89	J6201[28]	PRESS_INT	GPIO90
J6201[29]	MDP_VSYNC_S	GPIO83	J6201[30]	DISP1_RESET_N	GPIO25
J6201[31]	CAM4_RESET_N	GPIO116	J6201[32]	HALL_INT_N	GPIO80
J6201[33]	CAM5_RESET_N	GPIO115	J6201[34]	SAR_INT_N	GPIO91
J6201[35]	VREG_S10B_1P8	1.8V	J6201[36]	DC_IN	12V
J6201[37]	VREG_5P0	5.0V	J6201[38]	DC_IN	12V
J6201[39]	GND	ground	J6201[40]	GND	ground

3.4.16 On-board PCB WLAN antenna

The HDK8450 Development Kit has two on-board PCB antennas on the bottom side of the Main Board.

- CH0 is connected to WCN6856 by PCB trace
- CH1 is connected to WCN6856 via coaxial cable
- CH1 is also connected to GNSS
- For better GNSS performance, connect to external antenna via CH1_GNSS test connector

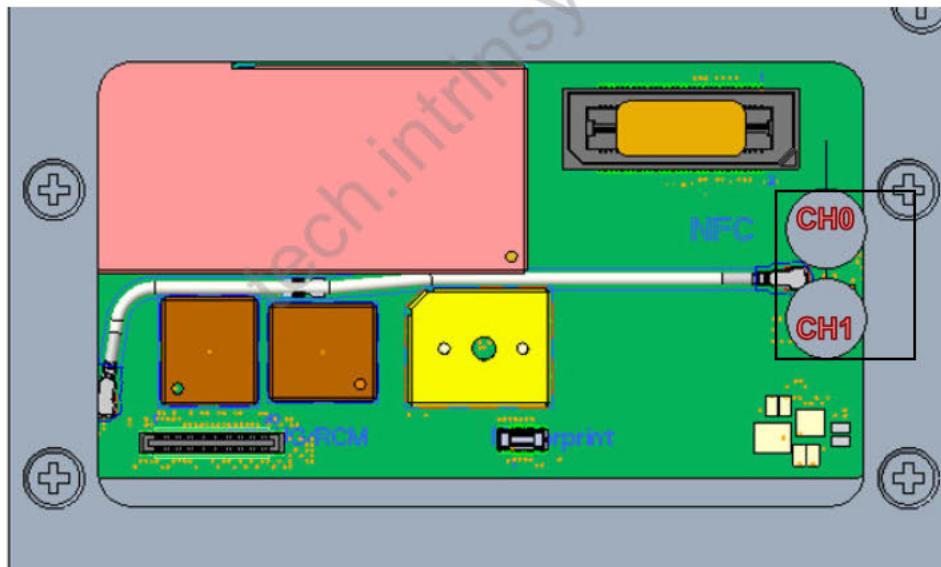


Figure 3-20 On-Board PCB WLAN Antennas

3.4.17 M.2 connector

The on board M.2 connector supports M/B card.



Figure 3-21 J6302 M.2 Connector

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4 Display card

The HDK8450 Display Card consists of the display panel and the Expansion Card. The supported features are:

- Copies of Power on, Vol+/- buttons of the Main Board.
- Dual MIPI DSI.
- Powers for OLED display and capacitive touch.
- I2C or SPI for Capacitive touch panel together with interrupts.
- 2 analog microphones interface on audio connector
- Sensor3 expansion connector for standard GenX.
- Sensor4 connector for legacy 24p sensor card.
- Jtag connector

4.1 Display card mechanical properties

Table 4-1 HDK8450 Expansion Card Mechanical Properties

Display	LCD 6.5' FHD 1080*2400 ROHS
Dimension	82.6cm ² (14.7 cm x 6.8 cm, max) – Expansion Card
Major Interfaces	2 x 60pin high speed and 1x40pin low speed board-to-board connectors

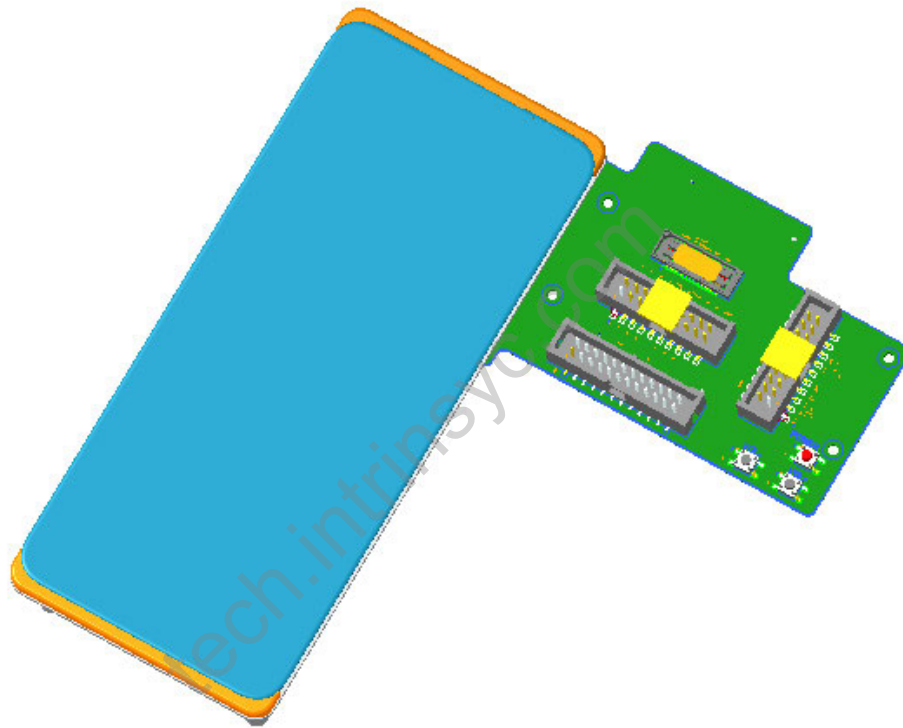


Figure 4-1 HDK Display Card

4.2 Expansion card connectors and features

Table 4-2 Expansion Card Connectors and Features

Domain	Description	Specification	Usage
Buttons	Power On	SMD Button	Red Power Button for Suspend / Resume and Power off
	Volume +	SMD Button	Volume + Key
	Volume –	SMD Button	Volume – Key
Audio connector	Audio/UART interface	2x Analog Microphone 1x MI2S 1x UART	Audio and UART expansion reserved for customer application
Jtag Connector	20-pin Jtag connectors	Standard 20-pin connector, support general JTAG	For general JTAG use

Domain	Description	Specification	Usage
Sensor3 expansion	60-pin GenX sensor connector	I3C, I2C, SPI, INTs, powers for sensor	For sensor gen board connection
Sensor4 connector	24-pin Legacy sensor connector	I3C, I2C, SPI, INTs, powers for sensor	For sensor cards used on legacy HDKs

4.2.1 Sensor3 expansion GenX connector

For dual I3C and I2C application, Sensor3 can be used with proper software setting.

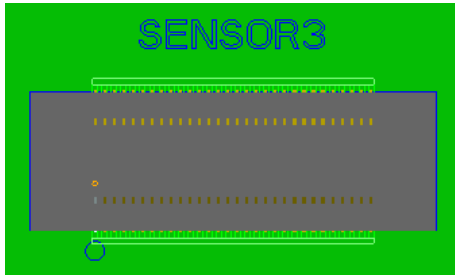


Figure 4-2 J701 Sensor3 Expansion GenX Connector

Table 4-3 Sensor3 Expansion GenX Connector Pinout

Pin#	Signal	Description	Pin#	Signal	Description
J701[1]	NC		J701[2]	NC	
J701[3]	NC		J701[4]	IMU_SPI_MOSI	GPIO 193
J701[5]	NC		J701[6]	IMU_SPI_MISO	GPIO 192
J701[7]	NC		J701[8]	IMU_SPI_SCLK	GPIO 194
J701[9]	NC		J701[10]	IMU_SPI_CS_N	GPIO 195
J701[11]	NC		J701[12]	NC	
J701[13]	NC		J701[14]	NC	
J701[15]	NC		J701[16]	NC	
J701[17]	NC		J701[18]	SAR_INT_N	GPIO63
J701[19]	NC		J701[20]	ALSP_INT_N	GPIO14
J701[21]	NC		J701[22]	HALL_INT_N	GPIO13
J701[23]	VREG_L8C_1P8_SNS	1.8V sensor digital power	J701[24]	NC	
J701[25]	NC		J701[26]	IMU2_INT	GPIO12
J701[27]	NC		J701[28]	PRESS_INT	GPIO83
J701[29]	NC		J701[30]	IMU1_INT	GPIO15
J701[31]	NC		J701[32]	MAG_INT_N	GPIO62
J701[33]	NC		J701[34]	SENSOR_I3C_SCL	GPIO 189
J701[35]	NC		J701[36]	SENSOR_I3C_SDA	GPIO 188
J701[37]	NC		J701[38]	VREG_L7C_3P0_NS	3.0V sensor analog power
J701[39]	NC		J701[40]	NC	
J701[41]	NC		J701[42]	NC	
J701[43]	NC		J701[44]	NC	

Pin#	Signal	Description	Pin#	Signal	Description
J701[45]	NC		J701[46]	NC	
J701[47]	NC		J701[48]	NC	
J701[49]	NC		J701[50]	NC	
J701[51]	NC		J701[52]	SENSOR_I2C_SDA	GPIO 196
J701[53]	NC		J701[54]	SENSOR_I2C_SCL	GPIO 197
J701[55]	NC		J701[56]	NC	
J701[57]	NC		J701[58]	NC	
J701[59]	NC		J701[60]	NC	

4.2.2 Sensor4 legacy sensor connector

Sensor4 is for HDK legacy sensor card support.

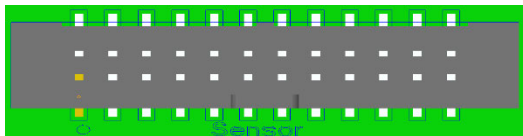


Figure 4-3 J702 Sensor4 Legacy Sensor Connector

Table 4-4 Sensor4 Expansion GenX Connector Pinout

Pin#	Signal	Description	Pin#	Signal	Description
J702[1]	SENSOR_I3C_SDA	GPIO188	J702[2]	IMU1_INT	GPIO15
J702[3]	SENSOR_I3C_SCL	GPIO189	J702[4]	PRESS_INT	GPIO83
J702[5]	NC		J702[6]	IMU2_INT	GPIO12
J702[7]	VREG_L8C_1P8_S NS	1.8V power supplier	J702[8]	VREG_L10C_3P0_S NS	3V power supplier
J702[9]	GND		J702[10]	GND	
J702[11]	SAR_INT_N	GPIO63	J702[12]	NC	
J702[13]	NC		J702[14]	ALSP_INT_N	GPIO14
J702[15]	NC		J702[16]	MAG_INT_N	GPIO62
J702[17]	NC		J702[18]	HALL_INT_N	GPIO13
J702[19]	IMU_SPI_CS_N	GPIO195	J702[20]	IMU_SPI_MOSI	GPIO193
J702[21]	IMU_SPI_CLK	GPIO194	J702[22]	IMU_SPI_MISO	GPIO192
J702[23]	NC		J702[24]	NC	

4.2.3 JTAG expansion connector

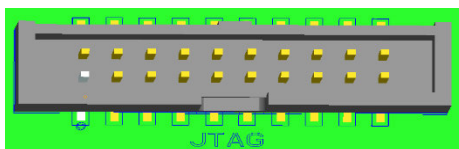


Figure 4-4 J601 JTAG Connector

This connector provides a JTAG interface to the main processor by which users can connect a JTAG (Lauterbach / USB Wiggler) 20 pin ARM JTAG.

NOTE: It does not provide software support for JTAG

Table 4-5 JTAG Header J601 Pin out

Pin NO	Signal	Description	Pin#	Signal	Description
J601[1]	JTAG_PWR	JTAG Power	J601[2]	NC	
J601[3]	TRST_N	Target RESET_N signal	J601[4]	DET_N	JTAG detect N signal
J601[5]	TDI	TDI Signal (Target DATA IN)	J601[6]	GND	GND
J601[7]	TMS	TMS Signal	J601[8]	GND	GND
J601[9]	TCK	TCK Signal	J601[10]	GND	GND
J601[11]	NC		J601[12]	GND	GND
J601[13]	TDO	TDO Signal (Target Data Out)	J601[14]	GND	GND
J601[15]	SRST_N	Source RESET_N signal	J601[16]	GND	GND
J601[17]	NC	NC	J601[18]	GND	GND
J601[19]	NC	NC	J601[20]	GND	GND

4.2.4 Audio expansion connector

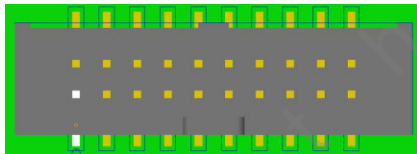


Figure 4-5 J501 Audio Expansion Connector

Pin NO	Signal	Description	Pin#	Signal	Description
J501[1]	EXP_UART_CTS	UART	J501[2]	MI2S2_I2S_WS	MI2S
J501[3]	EXP_UART_TX	UART	J501[4]	MI2S2_I2S_SCK	MI2S
J501[5]	EXP_UART_RX	UART	J501[6]	MI2S2_I2S_DAT0	MI2S
J501[7]	EXP_UART_RFR	UART	J501[8]	MI2S2_I2S_DAT1	MI2S
J501[9]	NC		J501[10]	WCD_MIC_BIAS3	Mic bias3
J501[11]	DC_IN	12V	J501[12]	WCD_MIC_BIAS1	Mic bias1
J501[13]	VREG_S10B_1P8	1.8V	J501[14]	DMIC01_DATA	GPIO_172
J501[15]	VREG_5P0	5V	J501[16]	DMIC01_CLK	GPIO_171
J501[17]	GND	Ground	J501[18]	DMIC23_DATA	GPIO_174
J501[19]	GND	ground	J501[20]	DMIC23_CLK	GPIO_173

4.3 Display panel

This display panel comes preinstalled on the HDK8450 expansion card. Below are the Panel specifications:

- Resolution: FHD 1080x2400 ROHS
- Display Type: LCD
- Capacitive touch panel
- Number of Lanes: 1x 3-tri MIPI DSI interface
- Diagonal Length: 6.5"

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5 Software notice

5.1 NV Items

Set below NV items to make the device work properly.

Table 5-1 NV items

NV#	Category	Value	Note
28967	RFNV_RF_CARD_CONFIG_I	913	The HWID for GNSS only config is 913.
00453	FactoryTest mode/Phone mode	0x00	Phone mode=0x00, FTM mode=0x01. Before measuring the rock bottom current, make sure NV00453 is set to 0x00, so that device can suspend/sleep.

5.2 CDT

- Platform ID info:

```
</device>
<device id="cdb0">
  <props name="platform_id" type="DALPROP_ATTR_TYPE_BYTE_SEQ">
    0x03, 0x1F, 0x01, 0x00, 0x01, 0x00, end
  </props>
</device>
```

- How to generate HDK8450 CDT bin:

```
Python boot_image/QcomPkg/Tools/cdt_generator.py
HDK8450_1.0.xml HDK8450_1.0.bin #update PLATFORM ID for HDK8450
```